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The Impact of Condensate Blockage and Completion Fluids on Gas Productivity in Gas-Condensate Reservoirs

H.A. Al-Anazi, SPE, J.R. Solares, SPE, and M.G. Al-Faifi, Saudi Aramco

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Abstract

Coreflood experiments were conducted on carbonate and sandstone cores from gas-condensate reservoirs in Saudi Arabia to assess the loss in gas relative permeability caused by condensate accumulation and water blockage. Field samples of condensate were used in these experiments to mimic two-phase flow around the wellbore region when the bottom hole flowing pressure dropped below the dewpoint. The impact of several fluids used as completion fluids was also investigated at reservoir conditions. Several solvents were evaluated to remove both condensate and water blockages.

Experimental results show that reductions of 70% to 95% in gas relative permeability were seen in reservoir cores due to condensate blockage. The studied solvents were found to be effective for enhancing gas relative permeability. This study also quantified the required methanol treatment volumes to increase gas relative permeability at lab conditions, which could be extrapolated to field conditions. The reduction in gas relative permeability was more pronounced during two-phase flow in the presence of water saturation due to the dual effect of condensate and water blockage. Methanol displaces retrograde condensate and maintains improved gas relative permeability well into the post-treatment production period.

Methanol-water mixtures were ineffective in removing condensate blockage and decreased gas productivity after their treatment. Methanol was effective in removing water from the cores. A mixture of isopropyl alcohol and methanol yielded similar favorable results as pure methanol. In summary, the evaluated solvents were all effective in removing condensate blockage from the core, delayed condensate accumulation, and enhanced gas productivity.

Introduction

Gas production from reservoirs flowing at bottom hole pressure lower than the dewpoint pressure, precipitates the accumulation of a liquid condensate in the near wellbore region. This condensate accumulation, also referred to as condensate banking, reduces the gas relative permeability and thus the well's productivity. Condensate saturations in the near wellbore region can reach as high as 50-60% under pseudo steady-state flow conditions.[1] Even when the gas is very lean, such as in the Arun field, with a maximum liquid drop out of 1.1%, condensate banking can cause a drastic decline in well productivity.[2-4] The Cal Canal field in California showed a very poor recovery of 10% of the original gas-in-place, because of the dual effect of condensate banking and high water saturation.[5]

Several methods have been proposed to restore gas production rates after a decline due to condensate and/or water blocking.[6-9] Gas cycling has been used to maintain reservoir pressure above the dewpoint. Injection of dry gas into a retrograde gas-condensate reservoir vaporizes condensate and increases its dewpoint pressure.[8] Injection of propane was experimentally found to decrease the dewpoint and vaporize condensate more efficiently than carbon dioxide.[10] Hydraulic fracturing has been used to enhance gas productivity, but is not always feasible or cost-effective.[5,11] Hydraulic fracturing is a commonly used technique to restore the gas productivity of wells in which the flowing bottom hole pressure has dropped below the dewpoint.[12]

High water saturation in the formation after a stimulation or workover treatment reduces the gas relative permeability. The adverse effect of condensate banking increases in the presence of high water saturation. A water block may occur when capillary forces exceed formation gas pressure. Under this scenario, water remains in the reservoir and it is flowed back at a very low rate. There are numerous examples of wells that required long periods of time to restore the initial gas productivity following liquid injection into the formation. The negative impact of water blockage increases in low permeability formations where the capillary forces are high or in low pressure reservoirs.[6,11,13]

Experimental tests involving the use of solvents to increase gas relative permeability reduced by condensate banking were conducted at lab conditions.[7,9,14] Methanol was found to be effective in removing both condensate and water and in restoring gas productivity in low permeability limestone cores. Al-Anazi et al.[9] studied the effect of condensate banking in both low and high permeability cores, and quantified the volumes of methanol required to reduce the effect. Gas productivity was found to decrease in similar fashion in both low and high permeability cores due to condensate banking. Order of magnitude productivity index increases, accompanied by condensate accumulation stoppage, were observed following a methanol treatment. Moreover, the productivity enhancement effect was maintained for an extended period of time regardless of permeability. The length of time at which the productivity enhancement effect is maintained is a function of the volume of methanol injected, and of the rate of mass transfer into the flowing gas phase after treatment.

A successful methanol treatment was implemented on a gas well in the Hatter's Pond field in Alabama.[15] The treated well showed a gradual decline in gas productivity and high skin prior to the treatment. The production history data suggested the possibility that condensate and water blocking had caused the productivity decline. Results after the treatment were very encouraging. Gas production increased by a factor of 3 initially and stabilized at over 500 MSCFPD, condensate production doubled to 157 BPD, and the skin was estimated at –1.9. The increase in gas and condensate production was sustained for more than 10 months, following the treatment, thus yielding long lasting benefits from methanol injection.

This paper discusses the results from a study conducted to ascertain the impact of condensate banking and completion fluids on gas productivity, and to evaluate the feasibility of treating wells with several solvents in order to restore the productivity.

Experimental Procedure

A coreflood apparatus was designed to mimic condensate banking in the near-wellbore region of wells flowing at below the dewpoint. **Figure 1** shows a schematic diagram of the coreflood apparatus. Pumps were used to inject fluids at constant flow rates at variable speeds up to 400 cc/min and pressure up to 10,000 psi. Accumulators with floating pistons rated up to 10,000 psi and 350°F were used to store gases and fluids. The core holder used can hold core plugs with a diameter of 1.5 inches and a length up to 3 inches. Pressure transducers were used to measure pressure drop across the core. The flow path was downward to eliminate gravity segregation effects. Two back-pressure regulators were used to control the flowing pressure upstream (BPR-1) and downstream (BPR-2) of the core. A visual cell was installed on-line downstream of the core to observe fluid phases at the effluent. The core holder, back-pressure regulators, fluid accumulators, and flow lines were inside a temperature-controlled forced-air circulation oven. The oven temperature is measured with a metal thermocouple and displayed on a digital indicator with an accuracy of ±1°C.

The core was loaded into the coreholder and the end-caps were tightened. As an overburden pressure of 4,000 psig was applied, the coreholder was placed inside the oven and kept there for 5 hours to ensure it reached the desired temperature. The pressure of the back-pressure regulators was adjusted to 1,500 psig.

The gas permeability was initially measured using nitrogen at a flowing pressure of 1,500 psig and 230°F (110°C). Nitrogen, condensate, and solvents were each stored in a separate rodded-piston accumulator. Each accumulator was placed vertically inside the oven and connected to a pump. In-situ two-phase flow conditions through the core were emulated by simultaneously flowing gas and condensate phases at a constant fractional flow factor. The injected phases were mixed together before they entered the upstream back-pressure regulator to create two-phase flow. Before the start of this stage, the inlet and outlet valves of the core were closed. The pressure of the upstream and downstream back-pressure regulators was set to 1,500 psig. Flow was started while the bypass valve was open until the pressure in the lines stabilized. Then, the bypass valve was closed and the inlet and outlet valves were opened simultaneously. This allowed flow of the nitrogen and condensate phases simultaneously through the core where it mimics the two-phase flow in the near-wellbore region. The flow was stopped when the pressure drop across the core reached stable values. A solvent (e.g. methanol, IPA, or methanol-water mixture) was injected through the core to remove the trapped condensate followed by a two-phase flow to assess the effectiveness of the solvent treatment.

Experiments were performed on cores from different gas wells whose properties are listed in **Table 1**. Condensate samples were collected from Well-G1 and Well-G2 and analyzed using a gas chromatography (GC). **Table 2** lists the compositional analysis of condensate field samples.

Results and Discussion

Condensate Blockage. Several two-phase flow tests were conducted using normal pentane (n-C5) and condensate field samples from Well-G1 and Well-G2. Cores from carbonate (Well-A and C) and sandstone (Well-B) reservoirs were used for the tests. Several of the reservoir and fluid characteristics which impact condensate banking, such as core permeability, critical condensate saturation, condensate composition, and initial water saturation were closely scrutinized.

Core Permeability. The effect of initial core permeability on the extent of condensate banking was explored using different cores with varying permeabilities. **Figure 2** shows the pressure drop across one of such cores during co-injection of nitrogen and n-C_5 at 1,500 psig and 230°F. The nitrogen and n-C_5 flow rates were 5 and 0.5 cc/min, respectively. The initial permeability for this particular core, at 100% gas saturation, was measured at 2.37 md. The liquid fractional flow of the flow rates was about 9% at injection conditions (32°F and 1,500 psig). The core fractional flow factor was estimated from flash calculations using the Peng-Robinson equation-of-state (PR-EOS). The pressure drop across the core gradually increased until it reached a maximum value of 24.5 psi. The condensate accumulated in the core until it reached ~~a~~ steady state flow conditions. The condensate phase broke through after 7.60 pore volumes (PV) of two-phase flow were circulated. Initially, the liquid phase filled up the pore space as the gas phase was flowing. When the pressure drop reached steady state conditions, both gas and liquid phases flowed at steady state fractional flow values. After condensate accumulation, the gas (k_{rg}) and oil (k_{ro}) relative permeabilities at steady state dropped to values of 0.191 and 0.117, respectively. This test showed that condensate accumulation during two-phase flow reduced gas relative permeability by 81%.

Another test was performed on a tight core from Well-B with an initial gas permeability of 0.26 md. **Figure 3** shows the pressure drop across the core during co-injection of nitrogen and n-C_5 at 1,500 psig and 230°F. Results from this test showed the same pressure build-up trend observed in the test performed on a more permeable core (**Figure 2**). However, this test required more pore volumes (17.84 PV) for the two-phase flow to reach steady state conditions, at which time the pressure drop across the core achieved a stabilized value of 193 psi. The fact that it took longer to achieve steady state conditions in a lower permeability core, even though the rate was kept constant throughout the tests involving high and low permeability cores, is explained by the higher displacement pressure needed for the low permeability core. In other words, the capillary forces in a low permeability core are expected to be higher than in a more permeable core. Condensate accumulation caused more than 80% reduction in gas productivity in this test.

The third coreflood experiment was conducted on a high permeability core from Well-B with an initial permeability of 17.28 md. Nitrogen and n-C_5 were co-injected through the core at the same flow rate, temperature, and pressure as in the other two tests in order to investigate condensate accumulation as a function of varying permeability. **Figure 4** depicts the pressure drop across the core during co-injection of nitrogen and n-C_5 at 1,500 psig and 230°F. Two-phase flow reached ~~a~~ steady state conditions after injecting 6.85 PV at which time the pressure drop stabilized at 3.76 psi. In this case the measured gas and oil relative permeabilities dropped to values of 0.270 and 0.164 respectively after condensate accumulation.

Figure 5 illustrates the drop in the gas productivity index observed after condensate accumulation in reservoir cores with varying permeabilities. The data clearly shows that, condensate banking reduces gas productivity regardless of permeability. However, the severity of productivity index (PI) reduction is higher in low permeability cores. For the sake of comparison, the post-condensate accumulation PI, measured in the different tests performed, was normalized to the pre-condensate accumulation PI and plotted in **Figure 6.** The data provide conclusive evidence that condensate banking decreased gas productivity in low and high permeability cores alike. These data support similar findings in previously conducted experiments.[9] In general, the PI of the different cores tested dropped more than 70% due to condensate banking.

Critical Condensate Saturation. The critical condensate saturation (S_{cc}) is defined as the minimum saturation required for the condensate phase to flow in a porous medium when the pressure is reduced below the dewpoint. It must be recognized that a degree of uncertainty is present in the measurement of the critical condensate saturation performed during these experiments because of the small pore volumes of the cores used. However, the critical condensate saturation can be predicted by applying a mass balance across the core at steady state conditions, and can be calculated using the following equation:

$$S_{cc} = f_o\,(1 + t_D) \qquad (1)$$

Where f_o is condensate fractional flow at steady state flowing conditions and t_D is pore volumes required to reach steady state at flowing conditions. Equation (1) is valid in the absence of initial water saturation. The condensate fractional flow at steady state conditions was estimated from flash calculations using the PR-EOS. Since both nitrogen and n-C_5 were co-injected using separate pumps set at room temperature and 1,500 psig, it was assumed that the fractional flow of n-C_5 would be smaller at a higher temperature of 230°F. The simulator estimated a fractional flow of 2.4 vol.% of n-C_5 at flowing conditions. However, this value is less than the 9.0 vol.% co-injected because a fraction of the n-C_5 evaporates at high temperature; therefore its fractional flow is lower at flowing conditions.

Table 3 lists the calculated critical condensate saturation for the different cores used in these experiments. These values show high critical condensate saturation in low permeability cores. This is due to higher capillary forces in low permeability cores, as indicated by the high pressure drop at steady state conditions and the increased condensate saturation observed in these cores. The critical condensate saturation values measured in these experiments are compare with those measured (24-51%) by Gravier et al.[16] who studied formation samples from a carbonate gas field with different permeabilities (0.4-50 md).

An interesting fact shown in Table 3 is that it took longer to reach the critical saturation of condensate accumulation in low permeability cores than in higher permeability ones. The implication is that the productivity decline resulting from condensate banking is expected to take longer to occur in low permeability reservoirs than in high permeability ones. However, condensate banking will ultimately reduce productivity in both types of reservoirs in severe fashion.[9]

Condensate Composition. The effect of condensate composition on condensate accumulation during two-phase flow was investigated comparing field condensate samples from Well-G1 and Well-G2 (**Table 2**) with n-C_5 on the same core. Each condensate phase was co-injected with nitrogen through the core, as two-phase flow stage, to assess the impact of condensate accumulation on gas relative permeability. Between each stage, the core was cleaned with methanol and flooded with large volume of nitrogen to restore its initial permeability. In each stage, nitrogen was injected simultaneously with the condensate at flow rates of 5 and 0.5 cc/min, respectively, at 1,500 psig and 230°F.

Figure 7 shows the gas PI profile during two-phase flow for various condensate samples. This result indicates that the gas PI decreased more with Well-G1 condensate than Well-G2 and n-C_5. Thus, gas relative permeability was reduced by 95% as given in **Table 4**. Condensate of Well-G1 has more heavy components (**Table 2**), high viscosity, and large interfacial tension (11.728 dynes/cm), as calculated by the parachor method with nitrogen than those of Well-G2 condensate and n-C_5. These factors magnify the capillary forces that reduce gas relative permeability more.

Residual Water Saturation. The effect of residual water saturation on gas and condensate relative permeabilities was investigated in reservoir cores using condensate field samples. The initial water saturation was established by flooding the core with reservoir brine (TDS = 199,455 ppm) followed by a water displacement with a nitrogen flood to get residual water saturation. **Figure 8** displays gas relative permeability measured during the displacement of brine by nitrogen flood at 1,500 psig and 230°F. The measured residual water saturation was found to be 34%. This value is high and expected to decrease by flooding more volumes of nitrogen. The single-phase gas (N_2) relative permeability was adversely decreased by water saturation. The presence of water reduces the available pore volume and consequently decreases gas permeability and the available pore space. The residual water saturation restricted condensate mobilization due to high capillary forces between the flowing gas phase and trapped water and condensate.[17]

Figure 9 shows the effect of residual water saturation on gas PI profile during two-phase flow. Both results were obtained from the same core using nitrogen and condensate from Well-G1. The presence of a residual water saturation resulted in a larger reduction in gas productivity during two-phase flow compared to that observed in a dry core. The same trend was observed when condensate of Well-G2 was used during two-phase flow as shown in **Figure 10**. **Table 5** gives the measured gas and condensate relative permeabilities during two-phase flow at different water saturations. The relative permeability to gas and condensate phases decreases more at high residual water saturation.

Solvents Treatment. Several solvents, such as methanol, methanol-water mixtures, isopropyl alcohol (IPA), and methanol-IPA mixtures were examined to assess their effectiveness in removing condensate blockage and enhance gas productivity.

Methanol. Methanol (100% pure) was injected through the core after condensate accumulation. The volume of the first methanol treatment was 20 PV. Then, two-phase flow was resumed directly after the treatment at the same fractional flow of the previous stage. **Figure 11** shows the PI profile during two-phase flow (N_2 and n-C_5) before and after methanol treatments at 1,500 psig and 230°F. This figure indicates that the methanol treatment increased the PI and delayed condensate blockage after the treatment. During methanol treatment, effluents of the core were collected in tubes using a fraction collector which was programmed to collect effluent each 5 minutes. **Figure 12** is a photo of the core effluent collected during methanol treatment. Methanol was effective in displacing the trapped condensate from the core as shown in the first four tubes that have yellowish color. Then, the effluent started to be become clear indicated a complete removal of condensate. When the two-phase flow resumed after the treatment, methanol rich phase in the pores prevents condensate accumulation and enhance gas productivity. **Table 6** gives gas and oil relative permeabilities measured before and after methanol treatments. The methanol treatments increased both gas and condensate relative permeabilities as a result gas productivity was enhanced as shown in **Figure 11**. Methanol treatment was evaluated using Well-G1 field sample as a condensate-phase in two-phase flow. The result is similar to that obtained with n-C_5 as shown in **Figure 13**, where the treatment enhanced the gas PI and delayed condensate accumulation. Therefore, gas and condensate relative permeabilities were increased by about 50% after the treatment as given in **Table 7**.

In case of using a condensate sample from Well-G2, the methanol treatment showed also an enhancement in gas PI that was increased almost by a factor of 2 as shown in **Figure 14**. Gas and condensate relative permeabilities were also increased after the methanol treatment as given in **Table 8**. As can be seen from the previous results, the methanol treatment was so effective in removing condensate blockage and enhancing gas productivity. The methanol is a temporary treatment that delays condensate accumulation due to the presence of methanol-rich phase. It is expected that condensate will starts to accumulate after stripping out all the methanol-rich phase from the core. The duration of methanol evaporation during two-phase flow will be discussed in the next section.

Methanol-Water Mixture. Methanol was found to be an effective solvent for removing condensate and water blockage and improving gas relative permeability.[7,9] However, it is flammable due to its low flash point (52°F) and high vapor pressure (4.63 psi at 100°F). To increase its flash point and minimize flammability risk, it can be mixed with water since it has an infinite miscibility in water. Thus, a mixture of low concentration of methanol in water has properties close to that of water. For instance, the surface tension of a water-methanol mixture has a value between that of water (72 dynes/cm) and methanol (22 dynes/cm) as shown in **Figure 15**.

Methanol-water mixture at concentrations of 25 and 50 vol.% methanol were examined as treatments after two-phase flow. **Figure 16** shows the PI profile during two-phase flow before and after treatments of methanol-water mixture at 1,500 psig and 230°F. These mixtures were not helpful in removing condensate from the core. However, treatment with 25 vol.% methanol had an adverse impact on the gas PI. This is because this mixture introduced water blockage besides condensate accumulation. As a result, the PI was decreased more than that the initial one measured during two-phase flow stage prior to the treatment. The mixture of 50 vol.% initially reduced the gas PI and showed gradual increase till it reached the initial PI value. This mixture removed the introduced water more than that with a lower methanol content (25 vol.%) because it has low surface tension and high vapor pressure. From practical point of view, pumping methanol-water mixtures will require a high injection pressure since they have higher viscosity than pure methanol.[14]

Mixing methanol in water will defeat its unique properties, such as low surface tension, low viscosity, high vapor pressure, and partial miscibility with condensate. Therefore, this mixture is not effective for removing condensate blockage and may decrease gas productivity after its treatment. Based on these results obtained, treatment with methanol-water mixtures is not recommended for gas-condensate reservoirs.

Isopropyl Alcohol. Isopropyl alcohol (IPA) was selected for investigation as an alternative for methanol treatment that gave promising results. Pure IPA and methanol-IPA mixtures were evaluated. **Figure 17** shows the gas PI profile measured during two-phase flow before and after treatments at 1,500 psig and 230°F. The examined solvents increased the gas PI almost to the same extent. **Table 9** summaries gas and condensate relative permeabilities measured during two-phase flow after various solvents treatments. These measured values show that methanol, IPA, or a mixture of both were all effective in removing the condensate from the core, delayed condensate accumulation, and resulted to enhancement in gas productivity.

Longevity of Methanol Treatment. Longevity of methanol treatment was investigated to assess the duration of the enhanced production period after the treatment. The core was treated with 2 stages of methanol treatment (20 PV each) separated by two-phase flow. The first treatment increased gas production by a factor of 1.5 during two-phase flow after the treatment as shown in **Figure 13**. It was decided to treat the core with the second methanol treatment and flood it with large volume of two-phase flow to determine the longevity of the treatment. Two-phase flow was injected continuously through the core for more than 3 days. **Figure 18** shows the effect of methanol treatments on gas PI at 1,500 psig and 230°F. As can be seen, the gas PI decreases gradually after the methanol treatment. This means that the stripping of methanol by the flowing gas-phase is a very slow process. After an injection of 577 PV of two-phase flow, the PI declined to a value close to that before the treatment. Therefore, the longevity of the treatment is quite long (577 PV) where the gas production can be increased and payback potential cost of the treatment.

Impact of Completion Fluids. Gas wells in tight reservoirs have shown low gas deliverability after drilling and completion operations. This is partially attributed to penetration of completion fluids used in these operations into the formation. The invaded fluids are trapped in the formation and form a ring around the wellbore, as a result, this high liquid saturation restricts or blocks the flow of gas. The increase in water saturation in the vicinity of the wellbore can play a significant role in the blocking of tight rock due to high capillary forces and vapor pressure. Water blocking is probably a significant issue in horizontal carbonate wells which may not be stimulated. The low initial drawdown in these wells keeps capillary trapped completion fluids in place for longer periods, therefore reducing productivity. Water blocking may also reduce gas productivity in fractured gas wells due to leakoff of fracture fluids into the fracture faces. Many fractured wells take a long time to clean-up completion fluids. Also of major concern is the potential for water blocking to affect the ability to restore well potential after mechanical workovers.

The impact of various completion fluids on gas relative permeability was explored on gas reservoir cores. **Table 10** lists the measured gas relative permeability after completion fluids and the cumulative gas volume injected. These completion fluids showed great adverse impact on gas relative permeability where gas productivity was declined. It required large volume of gas flood to clean some of the liquid from the core. However, mixing brine with alcohols gave better and faster cleaning as was mentioned previously.[6] Neat diesel gave lower gas relative permeability than brine. Diesel-water mixture resulted in more reduction probably due to emulsion formation which was observed in tube tests. Therefore, diesel is not recommended to be used as a completion fluid in gas wells with a perception of using lighter fluids.

Safety Issues on Methanol Treatment

Methanol has been widely used as a fracture fluid and an appropriate treatment fluid for stimulating water-sensitive formations.[19] However, the main drawback of methanol is its flammable nature due to its low flash point and high vapor pressure. Antoci et al.[11] reported that over 200 methanol jobs executed in Argentina without any safety incident. With the proper planning and following all safety guidelines, methanol can be pumped in such a safe manner to ensure that the job is implemented with minimal risk.

Conclusions

Condensate blockage caused a severe reduction in gas relative permeability and the extent of reduction increased in low permeability cores. The saturation of trapped condensate was predicted to be more in tight cores due to high capillary forces. Gas and condensate permeabilities were decreased more at high condensate saturation. Gas relative permeability reduced more during two-phase flow in the presence of water saturation due to the dual effect of condensate and water blockage. Methanol treatment was found to be effective for removing condensate blockage and enhancing gas productivity. The presence of methanol-rich phase in the core delayed condensate accumulation after the treatment. Thus, gas and condensate relative permeabilities were increased after methanol treatments during this delay duration.

Methanol-water mixtures were ineffective in removing condensate blockage and decreased gas productivity after their treatment. Methanol was effectual in removing water from the cores. Isopropyl alcohol and its mixture with methanol gave similar favorable results as pure methanol. These solvents were all effective in removing condensate blockage from the core, delayed condensate accumulation, and resulted to enhancement in gas productivity. The longevity of the methanol treatment was found to be quite long (more than 577 PV), where gas productivity was enhanced directly after the treatment and long lasting through it.

The examined completion fluids showed a low gas productivity due to high liquid saturation. The clean-up of these fluids is very slow. However, mixing brine with alcohols improved gas relative permeability.

Nomenclature

f_o = Fractional flow of condensate phase
k_i = Initial core permeability, md
k_{rg} = Gas phase relative permeability
k_{ro} = Condensate phase relative permeability
PI = Productivity index, cc/min/psi
PV = Pore volumes injected, dimensionless
S_{wr} = Residual water saturation, vol.%
S_{cc} = Critical condensate saturation, vol.%
t_D = Pore volumes at steady state, dimensionless
ΔP = Pressure drop, psi

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SI Metric Conversion Factors

cp	× 1.0*	E-03	=	Pa.s
ft	× 3.048*	E-01	=	m
°F	(°F-32)/1.8		=	°C
bbl	× 1.589873	E-01	=	m^3
in.	× 2.54*	E+00	=	cm
mL	× 1.0*	E+00	=	cm^3
psi	× 6.894757	E+00	=	kPa

*Conversion factor is exact.

Table 1: Properties of cores used in experimental studies.

	Well-A	Well-B	Well-B	Well-C
Lithology	Carbonate	Sandstone	Sandstone	Carbonate
Depth, ft	N/A	13,817.2	13,504.2	11,540.2
Length, in.	1.82	2.85	2.72	2.80
Diameter, in.	1.51	1.51	1.50	1.51
Bulk Volume, cm^3	53.41	83.64	78.67	82.17
Porosity, %	24.1	8.6	11.2	15.0
Pore Volume, cm^3	12.87	7.19	8.82	12.33
Permeability, md	1.20	0.26	17.28	1.54

Table 2: Compositional analysis of condensate field samples.

Component	Well-G1, mole%	Well-G2, mole%
Methane	0.00	0.63
Ethane	0.05	1.72
Propone	0.14	1.83
i-Butane	0.12	2.29
n-Butane	0.41	0.09
i-Pentane	0.73	4.09
n-Pentane	0.99	2.83
Hexanes	4.29	9.66
Heptanes	9.55	13.61
Octanes	13.51	17.13
Nonanes	12.81	12.21
Decanes	11.67	8.26
Undecanes	8.60	5.49
Dodecanes	7.60	4.43
Tridecanes	6.11	3.38
Tetradecanes	5.11	2.76
Pentadecanes	4.58	2.31
Hexadecanes	3.68	1.69
Heptadecanes	2.87	1.43
Octadecanes	2.83	1.23
Nonadecanes	2.43	1.04
Eicosanes	1.92	1.89
Molecular Weight	155.272	128.008

Table 3: Calculated critical condensate saturation in cores with various permeabilities.

k_i, md	t_D, PV	ΔP, psi	S_{cc}, %
0.26	17.84	192.98	45.22
1.54	8.18	30.00	22.03
2.37	7.6	25.55	20.64
17.28	6.85	3.63	18.84

Table 4: Gas and oil relative permeabilities measured during co-injection of nitrogen and condensate phase at 1,500 psig and 230°F.

Condensate	n-C_5	Well-G2	Well-G1
Viscosity, cp	0.1439	0.3950	0.5545
IFT*, dynes/cm	2.841	10.325	11.728
k_{rg}	0.270	0.088	0.051
k_{ro}	0.164	0.145	0.121

* Interfacial tension with nitrogen calculated by parachor method.

Table 5: Gas and oil relative permeabilities measured during two-phase at residual water saturation at 1,500 psig and 230°F.

Condensate	Well-G1	Well-G1	Well-G2	Well-G2
Gas Phase	N_2	N_2	N_2	N_2
S_{wr}, %	0	34	0	34
ΔP, psi	19.73	27.39	11.19	20.24
k_{rg}	0.051	0.035	0.116	0.064
k_{ro}	0.121	0.081	0.192	0.106

Table 6: Gas and oil relative permeabilities measured during two-phase flow (N_2 and n-C_5) before and after methanol treatments at 1,500 psig and 230°F.

	Before Treatment	After 1st Treatment	After 2nd Treatment
Treatment Volume, PV	0	13	15
k_{rg}	0.270	0.415	0.487
k_{ro}	0.164	0.253	0.297

Table 7: Gas and oil relative permeabilities measured during two-phase flow (N_2 and Well-G1 condensate) before and after methanol treatments at 1,500 psig and 230°F.

	Before Treatment	After 1st Treatment
Treatment Volume, PV	0	13
k_{rg}	0.051	0.075
k_{ro}	0.121	0.177

Table 8: Gas and oil relative permeabilities measured during two-phase flow (N_2 and Well-G2 condensate) before and after methanol treatments at 1,500 psig and 230°F.

	Before Treatment	After 1st Treatment
Treatment Volume, PV	0	15
k_{rg}	0.076	0.136
k_{ro}	0.126	0.225

Table 9: Gas and oil relative permeabilities measured during two-phase after several solvents treatments at 1,500 psig and 230°F.

	Before Treatment	After Methanol	After IPA	After MeOH-IPA (1:1)	After MeOH-IPA (1:3)
Treatment Volume, PV	15	15	15	15	15
k_{rg}	0.076	0.136	0.150	0.145	0.144
k_{ro}	0.126	0.225	0.248	0.240	0.238

IPA: isopropyl alcohol
MeOH: methanol

Table 10: Gas relative permeability measured after several completion fluids 1,500 psig and 230°F.

Completion Fluid	Gas Volume, PV	k_{rg}
Brine*	97.8	0.347
Brine + 10vol% MS**	70.9	0.395
7wt% KCl brine	99.7	0.402
100% Diesel	99.7	0.285
Diesel-Brine (3:1)	180.1	0.250
Diesel-Brine (1:1)	164.9	0.306
Diesel-Brine (1:3)	163.8	0.231
100% IPA	18.1	0.952
100% Methanol	18.9	1.000
MeOH-Brine(3:1)	34.1	0.890
MeOH-Brine(1:1)	41.5	0.726
MeOH-Brine(1:3)	52.7	0.426

*Mixing brine with TDS = 18,892 mg/L
** Mutual solvent

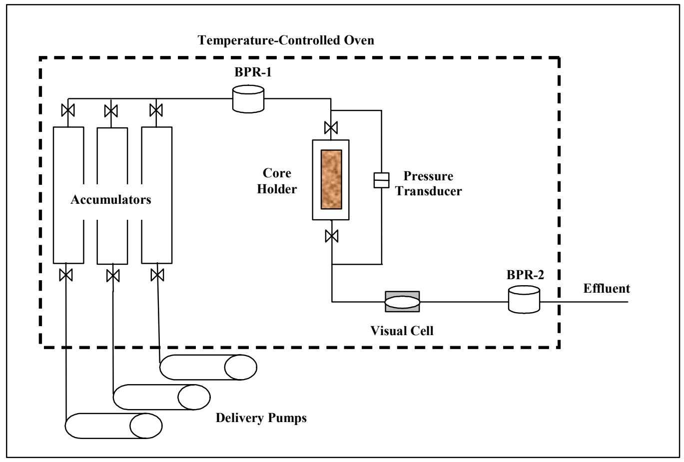

Figure 1: A schematic diagram of coreflood apparatus.

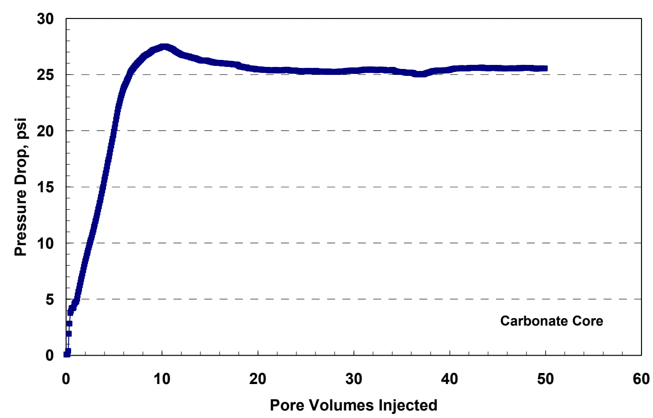

Figure 2: Pressure drop across the core during co-injection of nitrogen and n-C_5 at 1,500 psig and 230°F.

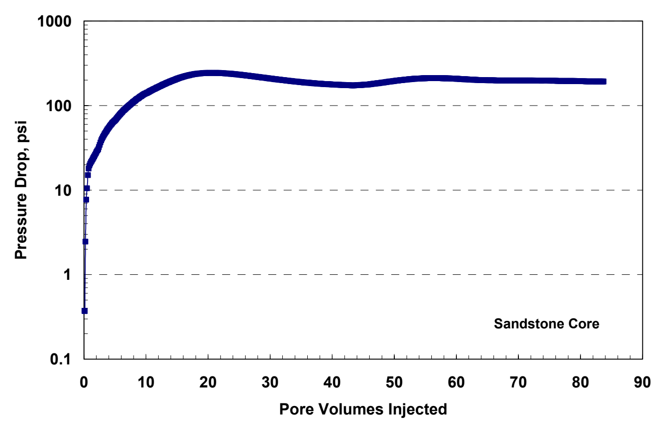

Figure 3: Pressure drop across the core during co-injection of nitrogen and n-C_5 at 1,500 psig and 230°F.

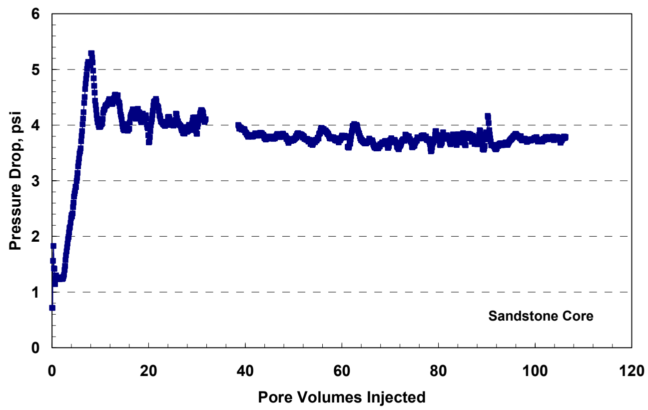

Figure 4: Pressure drop across the core during co-injection of nitrogen and n-C_5 at 1,500 psig and 230°F.

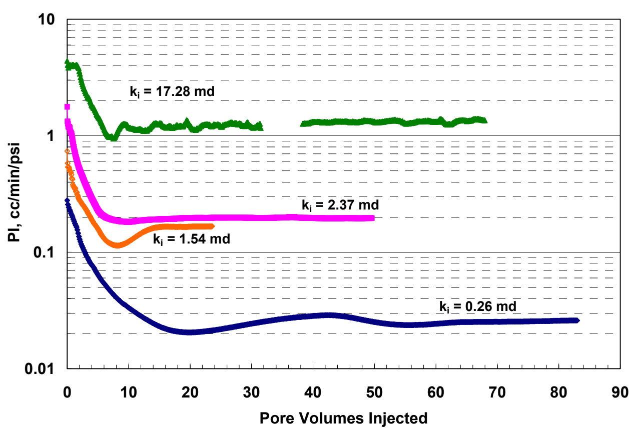

Figure 5: Productivity index profile during two-phase flow (N_2 and n-C_5) at 1,500 psig and 230°F.

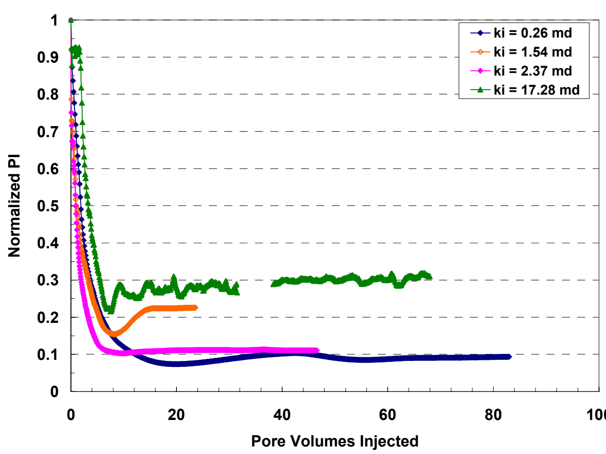

Figure 6: Normalized gas productivity index during condensate accumulation.

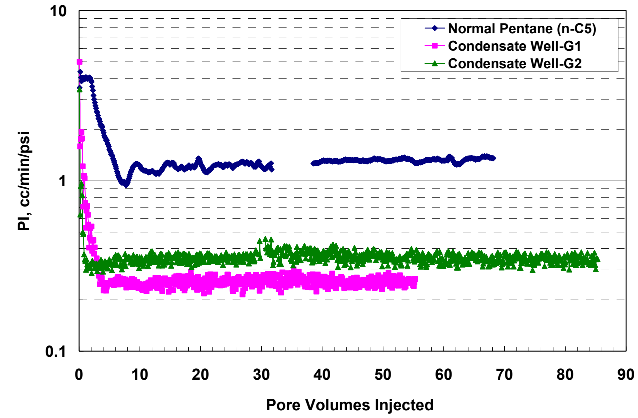

Figure 7: PI profile for different condensate phases during two-phase flow at 1,500 psig and 230°F.

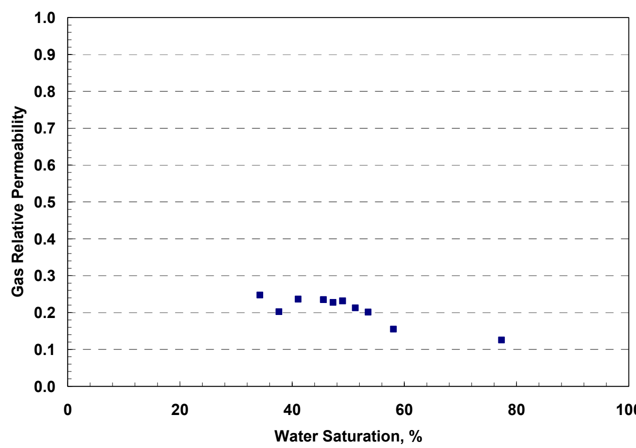

Figure 8: Gas relative permeability measured during brine displacement by nitrogen flood at 1,500 psig and 230°F.

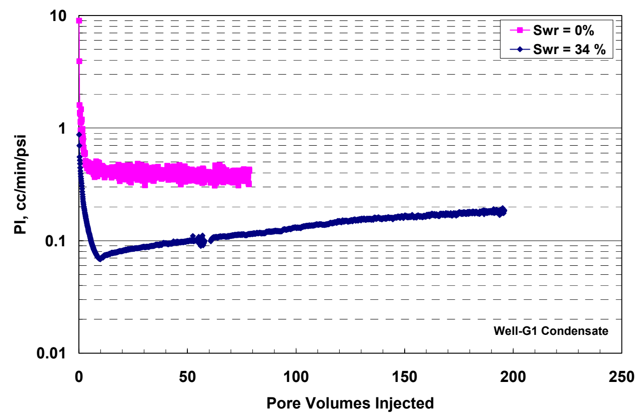

Figure 9: Effect of residual water saturation on gas PI profile during two-phase flow at 1,500 psig and 230°F.

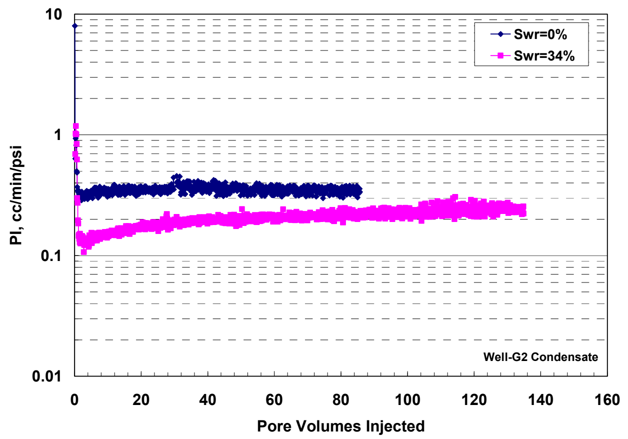

Figure 10: Effect of residual water saturation on gas PI profile during two-phase flow at 1,500 psig and 230°F.

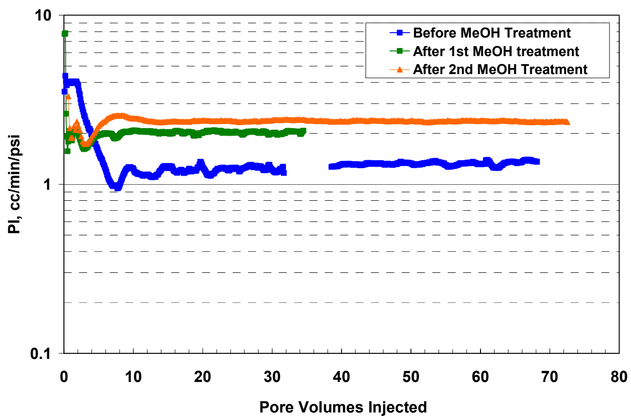

Figure 11: PI profile during two-phase flow (N_2 and n-C_5) before and after methanol treatments at 1,500 psig and 230°F.

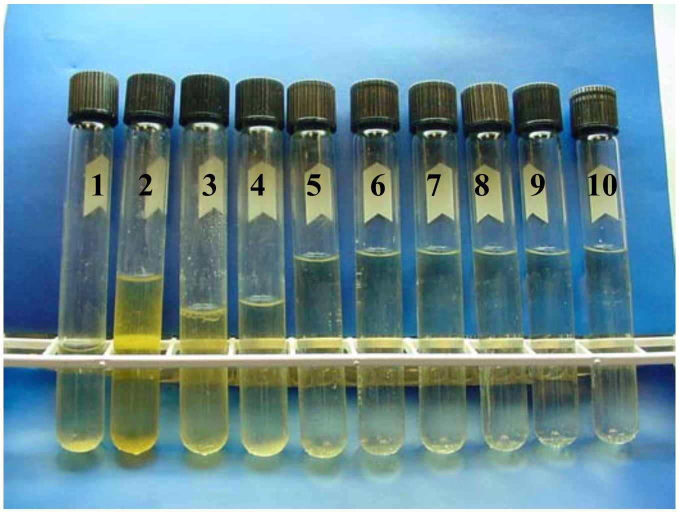

Figure 12: Effluent of the core during methanol treatment.

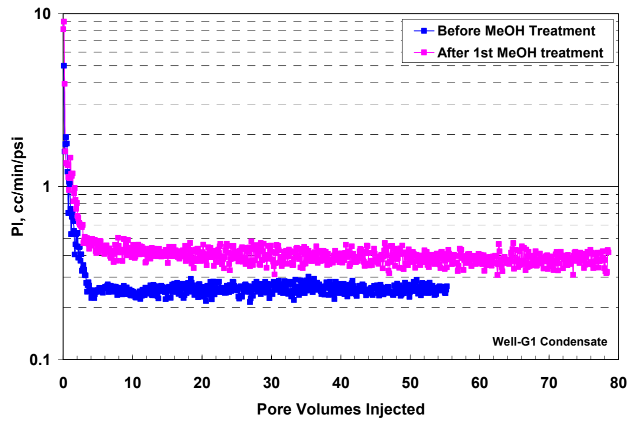

Figure 13: PI profile during two-phase flow (N_2 and Well-G1 condensate) before and after methanol treatments at 1,500 psig and 230°F.

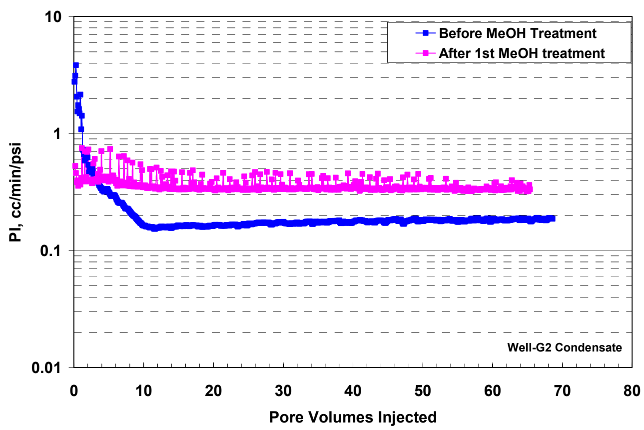

Figure 14: PI profile during two-phase flow (N_2 and Well-G2 condensate) before and after methanol treatments at 1,500 psig and 230°F.

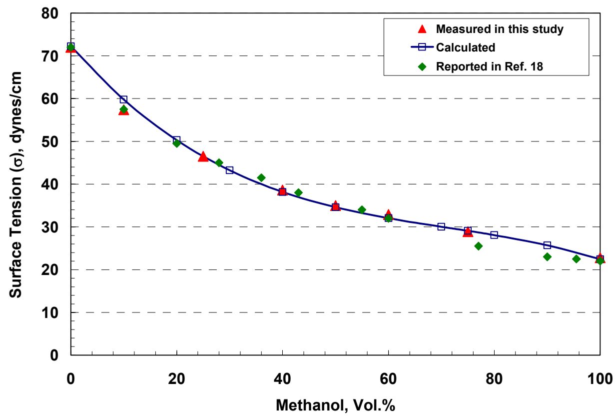

Figure 15: Surface tension of methanol-water mixtures at ambient conditions.

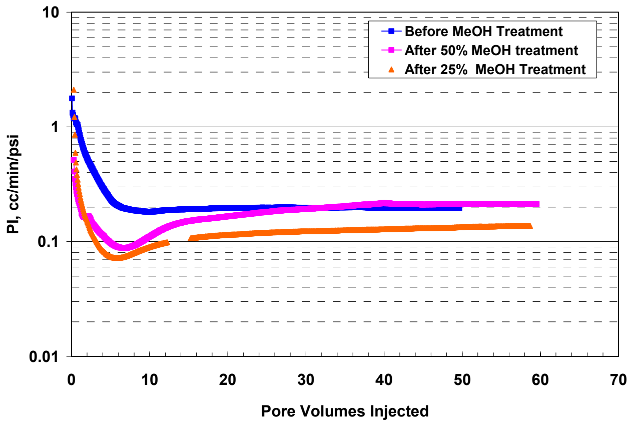

Figure 16: PI profile measured during two-phase flow before and after methanol-water mixture treatment at 1,500 psig and 230°F.

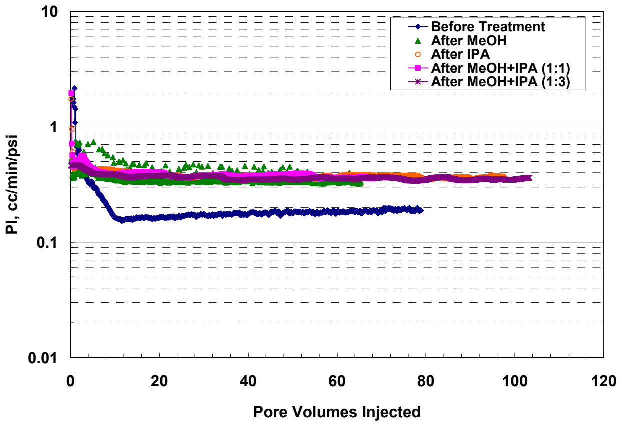

Figure 17: PI profile measured during two-phase flow before and after various solvent treatments at 1,500 psig and 230°F.

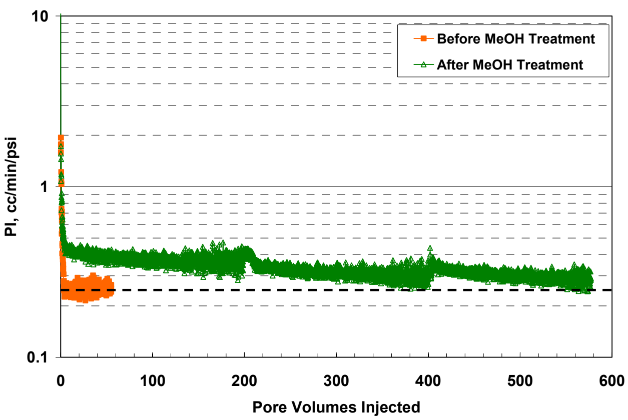

Figure 18: PI profile measured during two-phase flow before and after methanol treatment at 1,500 psig and 230°F to determine longevity of the treatment.